SATAR SHAMSI GOUSHKI

ML / AI Engineer

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Professional Summary

ML / AI Engineer covering machine learning, data engineering, and MLOps. Built an end-to-end LSTM forecasting system with MLflow and Optuna, a document intelligence pipeline (OCR, embeddings, vector search), ETL pipelines into PostgreSQL with Airflow, and FastAPI inference services deployed with Docker and CI. Completed an ML/AI engineering internship at GoalEarn. Graduate-level research at ITMO University in machine learning, image processing, and mathematical modeling. Open to remote and freelance work.

Technical Skills

Programming: Python, SQL, C++

AI & Machine Learning: Deep Learning, Neural Networks, LSTM, Natural Language Processing, Computer Vision, Classification, Time Series Forecasting, Hyperparameter Optimization

Frameworks & Libraries: TensorFlow, Keras, scikit-learn, NumPy, Pandas

NLP & Document AI: OCR, Embeddings, Sentence Transformers, Vector Databases, ChromaDB, LangChain

ML Engineering / MLOps: MLflow, Optuna, Model Tracking, Model Registry, FastAPI, Model Serving, Docker, GitHub Actions, CI/CD

Data Engineering: PostgreSQL, SQLAlchemy, ETL/ELT, Apache Airflow, Data Validation, Incremental Loading, Idempotent Loading

Development & Tools: Git, GitHub, Linux, Pytest, Structured Logging, HashiCorp Vault

Professional Experience

ML / AI Engineer Intern, GoalEarn – Iran — Oct 2025 – Sept 2026

- Built a modular ETL pipeline for acquiring, validating, transforming, and loading data into PostgreSQL, with incremental loading and structured logging.
- Added Pytest tests for pipeline components and used HashiCorp Vault for secure database credential management.
- Built an ML inference API serving model predictions through REST endpoints.

Selected Projects

LSTM Time Series Forecasting — End-to-End MLOps — Nov 2025 – present

Built an end-to-end machine learning system for time series forecasting, covering preprocessing, model development, evaluation, experiment tracking, inference, and deployment.

- Developed a stacked LSTM model with TensorFlow and Keras and evaluated it on a held-out test set.
- Used Optuna for hyperparameter optimization and MLflow for experiment tracking and model management.
- Identified and fixed a data leakage issue in the preprocessing pipeline that affected validation results.
- Built a FastAPI inference service and containerized the training and inference components with Docker.
- Added automated tests and GitHub Actions for CI/CD and deployed the inference API to Render.

Intelligent Document Analysis — July 2026 – Sept 2026

Built a document intelligence pipeline that extracts and prepares text from PDF and DOCX files for semantic search.

- Added PDF and DOCX processing using PyMuPDF and python-docx.
- Added OCR support for scanned documents using Tesseract.
- Built preprocessing workflows for extracting and preparing document text.

- Added embeddings and vector database support with Sentence Transformers and ChromaDB.
- Added automated tests for document ingestion and processing.

Market Data Pipeline — ETL

May 2026 – Sept 2026

Built a Python ETL pipeline that acquires market data from yfinance, validates and transforms it, and stores it in PostgreSQL.

- Implemented idempotent database loading and a pipeline-run audit table.
- Added data validation and 11 unit tests for the main pipeline components.
- Integrated Apache Airflow for scheduling and orchestration.
- Developed a FastAPI query layer for accessing stored market data.
- Used Docker to run the project services.

ML Inference API

Feb 2026

Built a REST API for serving machine learning model predictions.

- Developed prediction endpoints with FastAPI and request and response validation.
- Added environment-based configuration and structured logging.
- Added automated tests with Pytest for the main API components.
- Containerized the service with Docker for local development and deployment.

Retail Orders ETL Pipeline

Jan 2026 – Feb 2026

Built an ETL pipeline that loads validated retail order data from CSV files into PostgreSQL.

- Separated the pipeline into extraction, transformation, and loading stages.
- Added checks for missing values, negative amounts, duplicates, and required fields.
- Implemented incremental and idempotent database loading.
- Added structured logging and automated tests with Pytest.
- Used HashiCorp Vault for database credentials and containerized the pipeline with Docker.

Academic Research & Projects

Graduate Researcher — Big Data and Machine Learning, ITMO University – Saint Petersburg, Russia

Aug 2018 – May 2020

- Researched ML-based music generation; submitted a research paper.
- Researched and implemented an SEIR epidemic model.
- Worked on image classification and segmentation projects.
- Worked on parallel algorithms and HPC, including performance analysis.
- Completed projects in multivariate analysis, statistics, data warehousing, SQL, and MATLAB.

Education

ITMO University, Graduate Studies — Master's Program (degree not completed) in Big Data and Machine Learning – Saint Petersburg, Russia

Sept 2018 – May 2020

Completed coursework and research through the third semester.

- GPA: 3.91/4.00 in Big Data and Extreme Computing; 3.79/4.00 in Machine Learning and Data Analysis.

Shahid Bahonar University of Kerman, BSc in Electrical Engineering — Electronics – Kerman, Iran

Feb 2007 – Sept 2012

- GPA: 14.47/20.00 · Top 5 students

Awards

- Russian Open Doors Olympiad — Full Scholarship, HSE University (2024)

Languages

Spoken: English (professional), Persian (native), Russian (basic), German (basic)